

MONAHANS 6 ENE, TX, USA

WMO: 747290

Lat: 31.6219N

Lon: 102.8071W

Elev: 830

StdP: 91.74

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 03047

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.7	(c) -3.5	(d) -16.4	(e) 1.0	(f) 7.3	(g) -14.5	(h) 1.2	(i) 9.6	(j) 9.0	(k) 14.1	(l) 7.5	(m) 15.2	(n) 1.8	(o) N/A	(p) 0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	39.2	19.1	37.8	19.4	36.6	19.4	22.6	31.3	22.1	31.0	21.7	30.7	3.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	17.0	24.4	20.0	16.3	24.3	19.5	15.7	24.4	70.8	31.2	69.0	31.0	67.4	30.6	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.9	6.0	5.3		-9.8	41.6	2.4	1.4	-11.5	42.6	-12.9	43.4	-14.3	44.2	-16.0	45.2	
				DB												
				WB	-10.8	23.5	2.4	0.6	-12.5	23.9	-13.9	24.3	-15.2	24.6	-16.9	25.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.8	7.8	10.4	15.2	19.2	23.5	28.1	28.4	28.2	24.4	19.3	12.8	8.2
	DBStd	8.57	4.72	5.45	5.03	4.33	4.16	2.68	2.32	2.39	3.07	4.49	4.89	4.68
	HDD10.0	292	98	56	15	2	0	0	0	0	0	3	29	89
	HDD18.3	1252	327	224	120	41	10	0	0	0	2	43	173	313
	CDD10.0	3518	30	67	176	279	419	544	569	564	433	289	112	34
	CDD18.3	1437	0	2	23	68	171	294	311	306	185	72	6	0
	CDH23.3	17735	21	99	421	982	2142	3736	3848	3704	1862	768	127	24
	CDH26.7	9390	1	20	133	428	1141	2177	2190	2100	886	289	24	1
(14) Wind	WSAvg	2.5	2.3	2.6	2.8	2.9	2.9	3.0	2.7	2.4	2.3	2.2	2.1	2.2
(15) Precipitation	PrecAvg	361	14	17	17	16	36	34	41	47	47	34	16	17
	PrecMax	630	48	42	66	66	178	93	149	107	134	116	105	78
	PrecMin	202	1	0	1	0	0	1	0	1	0	1	0	0
	PrecStd	121	15	14	17	16	44	27	36	27	39	36	23	19
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.7	29.4	32.4	35.1	39.7	41.6	39.9	40.0	37.5	33.9	29.6	26.2
		MCWB	11.0	12.6	13.3	14.2	16.1	17.7	19.9	20.3	18.7	16.9	14.1	11.6
	2%	DB	23.1	26.3	30.1	33.4	37.1	39.7	38.2	38.3	35.2	31.9	26.7	23.2
		MCWB	10.0	11.7	12.8	14.0	15.6	18.1	20.2	20.1	19.1	17.1	13.7	10.8
	5%	DB	20.1	23.9	27.9	31.5	35.2	37.8	36.9	36.9	33.7	30.1	24.6	20.2
		MCWB	8.7	10.7	12.4	13.8	15.6	19.1	20.1	20.1	19.1	16.7	13.0	9.7
	10%	DB	17.2	20.9	25.6	29.2	33.0	36.0	35.5	35.4	31.8	27.9	22.1	17.2
		MCWB	7.6	9.6	12.0	13.7	16.2	19.7	20.1	20.3	19.0	16.5	12.0	8.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	15.5	17.6	19.0	21.9	23.1	23.7	23.0	22.7	21.8	18.0	14.2
		MCDB	20.5	22.1	25.4	26.2	29.5	32.4	30.2	32.1	30.4	29.4	22.5	18.7
	2%	WB	10.9	13.9	16.5	17.9	21.1	22.4	22.5	22.3	21.9	20.6	16.2	12.7
		MCDB	21.1	20.3	22.8	24.4	28.0	31.7	30.8	31.8	29.6	26.6	20.3	17.7
	5%	WB	9.5	12.3	15.3	17.0	20.4	21.8	22.0	21.8	21.4	19.5	15.1	11.4
		MCDB	18.7	20.9	21.8	23.6	26.9	31.1	30.8	31.6	28.8	25.2	20.4	18.0
	10%	WB	8.1	11.0	14.0	16.1	19.6	21.4	21.6	21.3	20.8	18.5	13.7	9.9
		MCDB	16.6	20.3	20.5	23.1	26.0	30.5	30.5	31.2	27.8	24.5	20.1	17.1
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	15.6	16.1	16.2	15.2	14.5	13.2	13.3	12.7	14.3	15.1	14.8
		MCDBR	20.4	20.2	19.6	19.3	18.7	17.0	15.2	15.2	15.6	16.9	18.1	19.8
		MCWBR	9.7	8.9	7.3	7.0	6.9	4.7	4.0	4.0	4.7	6.0	7.8	9.6
	5% WB	MCDBR	18.5	17.5	15.9	15.5	14.5	14.4	12.9	13.4	11.7	12.7	13.8	16.0
		MCWBR	9.0	8.4	6.3	6.5	6.4	4.3	3.5	3.8	3.8	5.2	7.1	8.6
(40) Clear-Sky Solar Irradiance	taub		0.263	0.272	0.291	0.326	0.349	0.377	0.399	0.371	0.353	0.307	0.284	0.268
	taud		2.583	2.561	2.516	2.402	2.397	2.374	2.351	2.413	2.432	2.555	2.548	2.570
	Ebn at Noon		981	999	996	963	937	906	884	908	915	944	944	957
	Edh at Noon		79	90	101	118	120	123	125	116	109	89	82	77
(44) All-Sky Solar Radiation	RadAvg		3.52	4.38	5.66	6.87	7.32	7.61	7.22	6.70	5.65	4.66	3.70	3.17
	RadStd		0.33	0.44	0.39	0.42	0.45	0.34	0.36	0.29	0.36	0.47	0.37	0.28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.02	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page